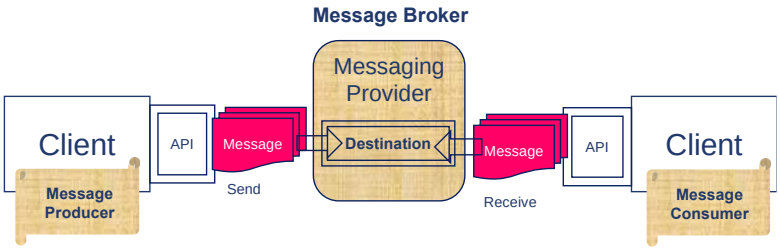
SOLUTIONS

Module:	Middleware		
Module Code	EBU6501	Paper	A
Time allowed	2hrs	Filename	**Solutions_2324_EBU6501_B**
Rubric	ANSWER ALL FOUR QUESTIONS		
Examiners	Dr Gokop Goteng	Dr Ethan Lau	

Question 1

a) Describe how the objects **`MessageProducer`** and **`MessageConsumer`** communicates using the concept of Message-oriented-middleware (MOM). Draw the MOM architecture diagram to aid the explanation.

(5 marks)

	Do not write in this column
Q1a) (Easy)	
MOM Architecture diagram:  [2 marks] 1. A **client (MessageProducer)** makes an **API** call to **send** a message. [1 mark] 2. Message sends to a **destination** managed by the provider. The call invokes provider services to route and deliver the message. [1 mark] 3. Once it has sent the message, the client **(MessageConsumer)** can continue to do other work, confident that the provider retains the message until a receiving **client retrieves** it. [1 mark]	
	5 marks

b) Apache ActiveMQ is an example of the Message Broker for the Java Message Service (JMS) application. It is used in publishing messages to a ***Topic*** or ***Queue***.

(15 marks)

i) Figures 1 (a-b) show the type of the JMS messaging model. Denote the type of the messaging model used in a) and b). Then, explain the difference of the two messaging models.

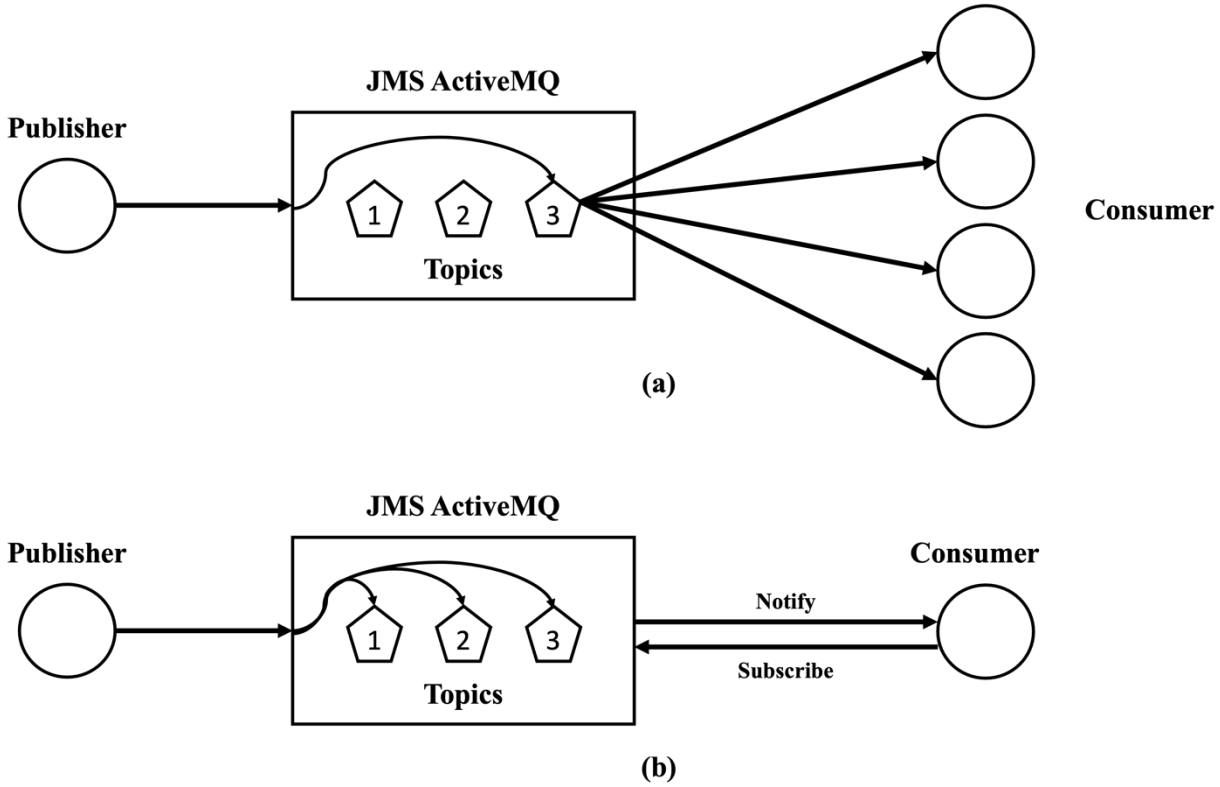


Figure 1 (a-b).

[6 marks]

ii) In your own words, describe the importance of the ***Sessions*** in the context of JMS Application Programming Interface (API) model.

[3 marks]

iii) What are the TWO important objects in JMS that are maintained administratively? Justify your answer accordingly.

[6 marks]

	Do not write in this column
Q1b)	
i) (Medium)	
a) is Topic-based model [1 mark]	
[NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] -It uses a hierarchical naming scheme [1 mark] -There is a logical arrangement in the way the topics appear using some form of classification method [1 mark] -It uses specific terms to describe the topics [1 mark] -It uses dedicated queues [1 mark] -Independent channels [1 mark] -Disadvantage - It is too restrictive as it uses specific terms [1 mark]	
b) is Content-based model [1 mark]	

[NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] -It does NOT use specific terms to describe the content [1 mark] -It uses commonly used query-like syntax for subscription of contents [1 mark] -Advantage: less restrictive as it does not use specific terms for subscriptions [1 mark] -(Un)structured queries [1 mark] -Complex Event Processing [1 mark]		
ii) (Medium)		
-A session is a single threaded context for producing and consuming messages [1 mark]. -Use sessions to create message producers, message consumers and messages [1 mark]. -Two forms: implement the **`QueueSession`** or the **`TopicSession`** interfaces [1 mark].		
iii) (Medium)		
The TWO Administered Objects in JMS and their descriptions are:		
Connection Factory (CF) [1 mark]: A connection factory is the object a client uses to create a connection with a provider [1 mark] which encapsulates connection configuration parameters [1 mark] **Destination (D)** [1 mark]**:** A destination is the object a client uses to specify the target of messages it produces [1 mark] and the source of message it [1 mark]		
		15 marks

c) The Inversion of Control (IoC) is a principle in the software engineering of the Spring Framework. It is most often used in the context of object-oriented programming such as Java.

(5 marks)

i) List out the TWO examples of the configuration metadata.

[2 marks]

ii) The Application Context is Spring's advanced container. Figure 2 shows a Spring's Application Context by instantiating the **`ApplicationContext`** interface that loads the .xml configurations from a classpath of "***info.xml***". Finally, the bean with the bean ID = **`beanInfo`** is obtained from the **`beanInfo`** class. Complete the missing information [i], [ii] and [iii] correspondingly.

[3 marks]

```
public static void main(String[] args) {

    SpringApplication.run(FirstSpringApplication.class, args);
    ClassPathXmlApplicationContext context=new [i]("[ii]");
    beanInfo sb=(beanInfo)context.[iii] ("beanInfo");
    sb.getName();
}
```

Figure 2.

	Do not write in this column
Q1c)	
i) (Easy)	
[NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] The configuration metadata can be represented either by; • XML, [1 mark]. • Java annotations, [1 mark]. • Java code. [1 mark].	
ii) (Easy)	
i) **`ClassPathXmlApplicationContext`** [1 mark] ii) ***Info.xml*** [1 mark] iii) **`getBean`** [1 mark].	
	5 marks

Question mark $\frac{}{5} + \frac{}{15} + \frac{}{5} = \frac{}{25}$

Question 2

a) Examine the Tomcat directory structure in Figure 3 below carefully.

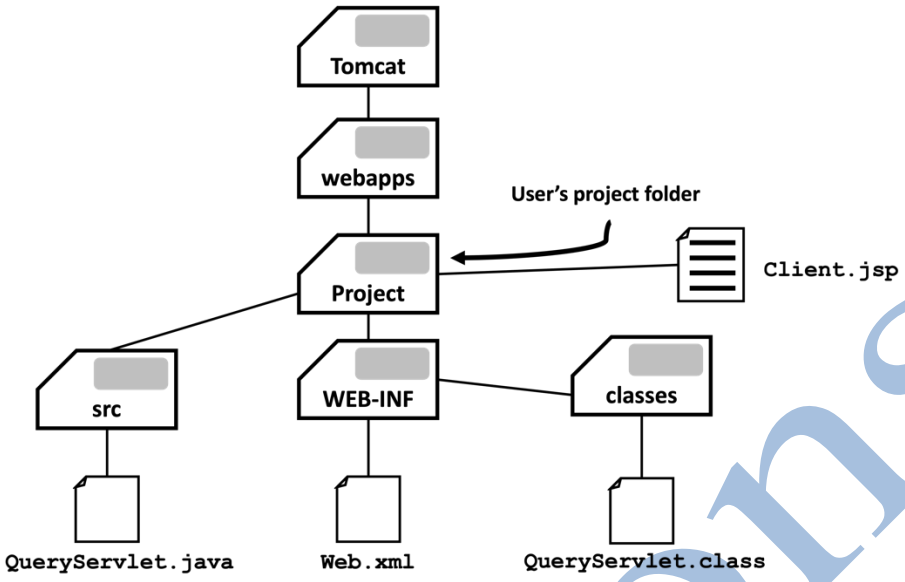


Figure 3.

(5 marks)

i) Define the term given to the file "**web.xml**" and explain why it is needed in the Tomcat Container.

[3 marks]

iii) Provide the correct URL using HTTP protocol and appropriate port number to access the web page "**`client.jsp`**" if you are running the system installed on your local machine.

[2 marks]

	Do not write in this column
Q2a)	
i) (Easy)	
-This XML file (web.xml) is called the Deployment Descriptor (DD) [1 mark]	
[NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] -It is used for setting up configurations [1 mark] -It is a configuration file for an artifact that is deployed to some container/engine [1 mark] -It is used for setting up initial parameters [1 mark]	
ii) (Easy)	
The correct URL is: **`http://localhost:8080/Project/Client.jsp`** or **`http://127.0.0.1:8080/Project/Client.jsp`** [2 marks]	
	5 marks

Solutions

b) The servlet class **`QueryServlet`** with servlet name **`Servlet1`** from Figure 4 is implemented in the Deployment Descriptor (DD) and the servlet class is in the directory **`com/query`** that extends **`HttpServlet`** class in Figure 4. The URL pattern is **`/result`**. Write servlet declaration in your DD for this servlet showing the complete package name of the class including the servlet-mapping and the URL in the DD.

```
public class QueryServlet extends HttpServlet {
      public void ModelDatabase() {
      System.out.print("<Query response result>");
      }
}
```

Figure 4.

(7 marks)

	Do not write in this column
Q2b) (Medium)	
<servlet> [0.5 mark]	
<servlet-name>Servlet1</servlet-name> [1 mark]	
<servlet-class>com.query.QueryServlet</servlet-class> [2 marks]	
[NOTE: <servlet-class> ModuleServlet</servlet-class> is acceptable [1 mark]]	
</servlet> [0.5 mark]	
<servlet-mapping> [0.5 mark]	
<servlet-name>Servlet1</servlet-name> [1 mark]	
<url-pattern>/result</url-pattern> [1 mark]	
</servlet-mapping> [0.5 mark]	
	7 marks

c) This question is about Java threads.

(13 marks)

i) Maintaining their order, state and explain the main FIVE life cycle states of a Java thread.

[6 marks]

ii) Figure 5 shows the Java class **`MyAccumulator`** where the method **getNextValue()** is not thread-safe as it is accessible by multiple threads. Improve the code in Figure 5 by acquiring an intrinsic lock to ensure the operation is thread-safe and atomic. Include the **`main()`** method to display the output result.

Solutions

[7 marks]

```
public class MyAccumulator {
  private int value = 50;
  public int getNextValue(){
            return value++;
``
```

Figure 5.

	Do not write in this column
Q2c)	
i) (Easy)	
FIVE states of the Java Thread are:	
New: This is when the Thread instance is created using the "new **`Thread()`**"command. [0.5 mark] The thread is still not alive yet [0.5 mark]	
Runnable: This is when the Thread instance variable invokes the "start()" method using the "**`t.start()`**" command, where **`t`** is the thread instance variable [1 mark]. The thread is now alive and has joined the queue of threads about to be executed [1 mark]	
Running: This is when the Thread is now executing [1 mark]	
Blocked/Waiting/Sleeping: This is when the Thread goes out of the execution or running mode when the wait, interrupt or sleep commands are executed [1 mark]	
Dead: This is when the Thread completes its task terminates execution calling "**`t.destroy()`**" command for garbage collection [1 mark]	
ii) (Hard)	
`public class MySubtraction {`	
`private int value = 10;`	
//[Note: either one of the method (Method 1 or 2) is acceptable [**3 marks** maximum for each of the method]]	
`// Method 1`	
`public int getNextValue()` [1 mark] `{`	
`synchronized (this)` [2 marks] `{ return value--;` [1 mark] `}}`	
`// Method 2`	
`public synchronized` [2 marks] `int getNextValue()` [1 mark] `{ return value--;` [1 mark] `}}`	
`public static void main(String[] args) {` [1 mark]	
`MyCalculation e =new MyCalculation ();` [1 mark]	

Solutions

`System.out.println(e.getNextValue());}` [1 mark] *		
`}`		
* **Note:** Other forms of output display are acceptable.		
		13 marks

Question marking $\frac{}{5} + \frac{}{7} + \frac{}{13} = \frac{}{25}$

Solutions

Question 3

a) This question is on Apache Kafka:

[16 marks]

i) Describe how ***Apache Kafka Streams*** simplifies middleware application development.

(8 marks)

ii) Describe how ***Apache Kafka Streams*** are prepared for storing, transporting and processing them.

(5 marks)

iii) Describe how ***Apache Kafka Streams*** enables its cluster to have data locality, elasticity, scalability, high performance and fault tolerance system.

(3 marks)

	Do not write in this column
Q3a i)	
Apache Kafka Streams simplifies application development by building [1 mark] on the Kafka producer [1 mark] and consumer libraries [1 mark] and leveraging the native capabilities [1 mark] of Kafka to offer data parallelism [1 mark], distributed coordination [1 mark], fault tolerance [1 mark], and operational simplicity [1 mark].	
Q3a ii)	
The messaging layer [1 mark] of Kafka partitions data [1 mark] for storing [1 mark] and transporting it [1 mark].	
Kafka Streams partitions data for processing it [1 mark].	
Q3a iii)	
Because of the Kafka data layer and streams partitioning [1 mark], it enables data locality, elasticity, scalability [1 mark], high performance, and fault tolerance [1 mark].	
Note: students can also use different words to explain the same concepts.	
	16 marks

b) Describe ***Injection Flaws*** and ***Cross Site Request Forgery (CSRF)*** security problems and how to prevent each of them.

[7 marks]

	Do not write in this column
Q3b)	
Injection Flaws: Injection flaws result from failure to filter untrusted input [1 mark]. It	

can happen when you pass unfiltered data to the SQL server (SQL injection), to the Lightweight Directory Access Protocol (LDAP) server (LDAPInjection), etc [1 mark]. The attacker can "inject" commands to these entities, resulting in loss of data and hijacking clients' browsers. [1 mark]	
You can prevent Injection Flaws by adopting highly skilful programming and encryption techniques plus vigorous testing procedures and Updating browsers regularly [1 mark].	
Site Request Forgery (CSRF): CSRF is caused by a third party browser that is not authentic can deceive you to enter sensitive details (for example your banker's site may be compromised and you may enter your details which may be available to the attacker) [1 mark]. This is sometimes called the "Confused Deputy" problem. [1 mark]	
You can prevent CSRF by avoiding clicking on URLs that are suspicious [1 mark].	
	7 marks

c) Describe ONE IoT constrained protocol that is used at the application layer for sensors.

[2 marks]

	Do not write in this column
Q3c CoAP (Constrained Application Protocol): CoAP is an application layer and web-based protocol designed for constrained devices like sensors. [1 mark]	
CoAP is similar to the HyperText Transport Protocol (HTTP) protocol an is used as the Representative State Transfer (REST) architecture. [1 mark]	
	2 marks

Question marking: $\frac{}{16} + \frac{}{7} + \frac{}{2} = \frac{}{25}$

Solutions

Question 4

a) A company intends to use a secure application programming interface (API) registry management service that allows bundles to be remotely managed for developing middleware using Java technology. Describe THREE features in OSGi (Open Service Gateway Initiative) that can meet these requirements.

[6 marks]

	Do not write in this column
Q4a	
Security (secure means security feature) [1 mark]**:** The security layer that handles authentication, authorisation and protection of bundles [1 mark].	
Service Registry (API for registering/managing services) [1 mark]**:** This is the Application Programming Interface (API) for management services that register services and tracks down services based on their unique service references [1 mark].	
Lifecycle (API to manage bundles remotely) [1 mark]**:** The application programming interface (API) that allows bundles to be remotely installed, stopped, started, updated, uninstalled on the fly without restarting the services [1 mark].	
	6 marks

b) A company wants to put in place a security system that ensures secure way for users to login, have access to the computer system and use the services. In addition, it wants to ensure that data is not seen by unauthorised users and that data cannot be manipulated while stored on the system or in transit. Discuss FOUR ways that you can help this company to achieve these objectives.

[8 marks]

	Do not write in this column
Q4b)	
Authentication [1 mark]**:** This is a process that you can use username and passwords or other means to allow users to login the system of the company [1 mark].	
Authorisation [1 mark]**:** This feature consists of policies that allow users that have login the system to be able to user the services in the company's computer systems [1 mark].	
Data Integrity [1 mark]**:** This is the ability of computer systems to ensure that data at rest and in transit is not changed or manipulated by unauthorised users and that the originality of data must be preserved without anybody tempering with it in an unauthorised way [1 mark].	
Data Confidentiality [1 mark]**:** This is the ability of computer systems to ensure that data can only be made available or accessible or seen by only the intended audience or	

user of the data, meaning, data should not find its way into the wrong hands [1 mark].		
		8 marks

c) Describe the FOUR steps in using the Uniform Resource Locator (URL) to write and implement a Java networking program.

[11 marks]

	Do not write in this column
Q4c	
Step 1: Create a URL [2 marks] Within your Java programs, you can create a URL object that represents a URL address which always refers to an absolute URL but can be constructed from an absolute URL, a relative URL or from URL components. [1 mark]	
Step 2: Reading from a URL [2 marks] You can read from a URL using the openStream() method. [1 mark]	
Step 3: Connecting to a URL [2 marks] You can connect to URL by calling openConnection() method on the URL which returns a URLConnection object that you can use for more general communications with the URL. [1 mark]	
Step 4: Reading from and Writing to a URL Connection [1 mark] This ensures how to connect to a URL and get results back. [1 mark]	
	11 marks

Question marking $\frac{}{6} + \frac{}{8} + \frac{}{11} = \frac{}{25}$

Solutions